Generalized linear models

Hypotheses testing

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Hypotheses testing - Contents

Testing problem

- Often one is interested in statements on single coefficients
- Example: Does covariate x_j have an effect on y? $\rightarrow$ Test on β_j
- Consider general testing problem with linear hypotheses

$$\begin{aligned} H_o: &\quad \mathbf{C}\beta = \mathbf{d} \\ \text{vs. } H_1: &\quad \mathbf{C}\beta \neq \mathbf{d} \end{aligned}$$

with $\mathbf{C} \in \mathbb{R}^{r\times(p+1)}$, $\boldsymbol{\beta} \in \mathbb{R}^{p+1}$, $\mathbf{d} \in \mathbb{R}^r$, whereas $r \leq p+1$ und $\text{rang}(\mathbf{C}) = r$ ($\Longrightarrow$ **C** is of full row rank).

Testing problem

Examples

(1) $H_0: \beta_1 = 0$ vs. $H_1: \beta_1 \neq 0$

$$(0,1,0,\dots,0)\cdot\begin{pmatrix}\beta_0\\ \beta_1\\ \vdots\\ \beta_p\end{pmatrix} = 0$$

(2) $H_0: \beta_3 = \beta_4$ vs. $H_1: \beta_3 \neq \beta_4$

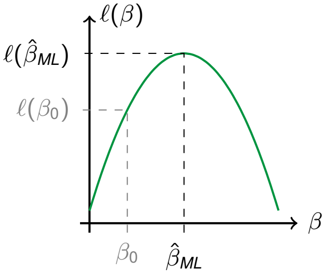
$$(0,0,0,1,-1,0,\dots,0)\cdot\begin{pmatrix}\beta_0\\ \beta_1\\ \vdots\\ \beta_p\end{pmatrix} = 0$$

$$\Longleftrightarrow \beta_3 - \beta_4 = 0$$

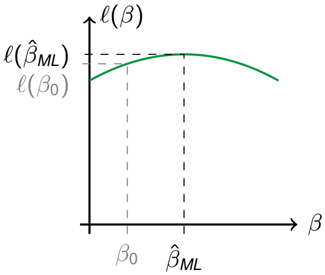
Likelihood-ratio test

Idea:
Compare (Log-)Likelihood on the conventional ML estimator $\hat{\beta}_{ML}$ (no restrictions) and on the ML estimator β_0 under the hypothesis H_0 (restriction on $\mathbf{C}\beta = \mathbf{d}$).

Example: $p = r = 1: \quad H_0 : \beta = \beta_0 \quad \text{vs.} \quad H_1 : \beta \neq \beta_0$



$\ell(\hat{\beta}_{ML}) > \ell(\beta_0)$
reject H_0



$\ell(\hat{\beta}_{ML}) \approx \ell(\beta_0)$
retain H_0

Likelihood-ratio test

The Likelihood-ratio (LR) is defined via

possible test statistic

$$\mathrm{LR} = \frac{\sup\limits_{\beta \text{ fullfils } H_0} L(\beta)}{\sup\limits_{\beta \text{ fullfils } H_0 \text{ or } H_1} L(\beta)} = \frac{L\left(\tilde{\beta}\right)}{L\left(\hat{\beta}\right)}$$

restricted

unrestricted

with

- $\tilde{\beta} =$ ML-estimate in the restricted model
- $\hat{\beta} =$ — " — unrestricted —"—

Likelihood-ratio test

- Restricted model is a special case of the unrestricted one
- Therefore:

$$L(\hat{\beta}) \geq L(\tilde{\beta}) \quad ,$$

also

$$\mathrm{LR} \leq 1 \quad .$$

- Interpretation:
  - $\mathrm{LR} \ll 1 \quad \rightarrow$ argues against H_0
  - $\mathrm{LR} \approx 1 \quad \rightarrow$ H_0 is no restriction, H_0 could be correct

LR is sensible for our testing problem

Likelihood-ratio test

test statistic

Log-likelihood ratio statistic

$$\mathbb{R}_+^0 \ni \quad \lambda = -2\log(\mathrm{LR}) = -2\left(l(\tilde{\beta}) - l(\hat{\beta})\right)$$

Interpretation:

still sensible

- Large λ: argues against H_0 ($\log(0.1) = -2.3$)
- Small λ: argues pro H_0 ($\log(1) = 0$)

Asymptotics (under H_0):

$$\lambda \overset{a}{\sim} \chi^2_r$$

Likelihood-ratio test

How can λ be computed in practice?

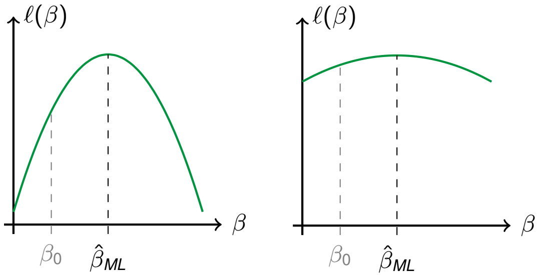
- $l(\hat{\beta})$: ML-estimator (use usual)
- $l(\tilde{\beta})$:
  - optimization under side constraints
  - only easy in special cases, e.g. excluding a certain covariate

$\Rightarrow$ typically cumbersome

Wald test

Idea:
Under H_0 we have $\mathbf{C}\beta - \mathbf{d} = \mathbf{0}$. Regard the (squared) distance $\mathbf{C}\hat{\beta}_{ML} - \mathbf{d}$, but also considering the curvature.

Example: $H_0 : \beta = \beta_0 \quad \text{vs.} \quad H_1 : \beta \neq \beta_0$



Strong curvature, reject H_0

Little curvature, retain H_0

Wald test

Idea:

Regard the difference between $\mathbf{C}\hat{\beta}$ and $\mathbf{d}$ via the norm

$$||\mathbf{C}\hat{\beta} - \mathbf{d}||_2^2 = (\mathbf{C}\hat{\beta} - \mathbf{d})^\top (\mathbf{C}\hat{\beta} - \mathbf{d})$$

sensible for our testing problem

Interpretation:

- If distance is small, this supports H_0
- If distance is large, this argues agains H_0

$\hat{\beta}$ random variable $\rightarrow$ distance also random
Use weights, namely the inverse of

$$\mathrm{Cov}(\mathbf{C}\hat{\beta} - \mathbf{d}) = \mathbf{C}\,\mathrm{Cov}(\hat{\beta})\mathbf{C}^\top \approx \mathbf{C}\,\mathbf{F}(\hat{\beta})^{-1}\mathbf{C}^\top \,.$$

curvature

Wald test

$$\underline{A}^{\frac{T}{2}} \, \underline{A}^{\frac{1}{2}} = \underline{A}$$

Wald statistic

$$
\begin{aligned}
w &= (\mathbf{C}\hat{\beta} - \mathbf{d})^\top \Big(\overbrace{\mathbf{C}\mathbf{F}(\hat{\beta})^{-1}\mathbf{C}^\top}^{\operatorname{cov}(\underline{C}\hat{\beta})}\Big)^{-1} (\mathbf{C}\hat{\beta} - \mathbf{d}) \\
&\approx (\mathbf{C}\hat{\beta} - \mathbf{d})^\top \operatorname{Cov}(\mathbf{C}\hat{\beta})^{-\frac{1}{2}} \operatorname{Cov}(\mathbf{C}\hat{\beta})^{-\frac{1}{2}} (\mathbf{C}\hat{\beta} - \mathbf{d}) \\
&= \Big[\operatorname{Cov}(\mathbf{C}\hat{\beta})^{-\frac{1}{2}} (\mathbf{C}\hat{\beta} - \mathbf{d})\Big]^\top \Big[\operatorname{Cov}(\mathbf{C}\hat{\beta})^{-\frac{1}{2}} (\mathbf{C}\hat{\beta} - \mathbf{d})\Big]
\end{aligned}
$$

Asymptotics (under H_0):

$$w \overset{a}{\sim} \chi^2_r$$

Wald test

Benefit of the Wald test towards LR:

- No estimation of the restricted model necessary!
- Can be determined simultaneously for various hypotheses
- Overall, it is simpler and therefore implemented in programming packages

$\tilde{\beta}$ not needed

Wald test

- **Special case**

  $H_0 : \beta_s = 0$ vs. $H_1 : \beta_s \neq 0$ for $s \in \{1, ..., p\}$
  $\implies \mathbf{C} = (0, ...0, 1, 0, ..., 0)$

  !

  $$\implies w = (\hat{\beta}_s - 0)\frac{1}{a_{ss}}(\hat{\beta}_s - 0)\,,$$

  where a_{ss} is the s-th diagonal element of $\mathbf{F}(\hat{\beta})^{-1}$, because $\frac{1}{a_{ss}} = \left(\mathbf{C}\mathbf{F}(\hat{\beta})^{-1}\mathbf{C}^\top\right)^{-1}$

  $$\implies w = \left(\frac{\hat{\beta}_s}{\sqrt{a_{ss}}}\right)^2 \underset{H_0}{\overset{a}{\sim}} \chi^2_1$$

  $X_i^2 \sim \chi^2_1$

  $\Updownarrow$

  $X_i \sim N(0,1)$

  and

  $= 1$ if $\hat{\beta}_s > 0$
  $= -1$, if $\hat{\beta}_s < 0$

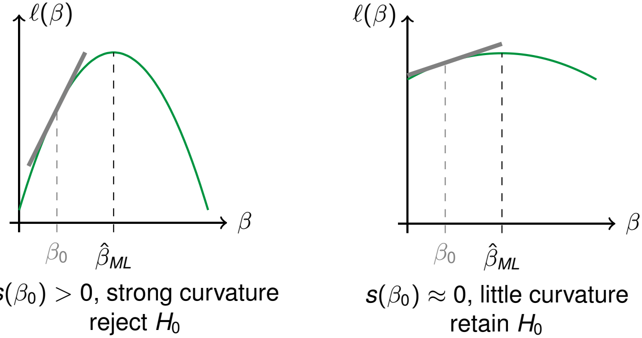
  $$\implies \text{sign}(\hat{\beta}_s)\sqrt{w} = \frac{\hat{\beta}_s}{\sqrt{a_{ss}}} \underset{H_0}{\overset{a}{\sim}} N(0, 1)\,.$$

  (appears e.g. as `z-value` in software output.)

Score test

Idea:
For the ML estimator (without restrictions) $\mathbf{s}(\hat{\beta}_{ML}) = \mathbf{0}$ holds.
Likewise, under H_0, for the restricted estimator $\mathbf{s}(\beta_0) \approx \mathbf{0}$ should hold. Here, it is weighted with the curvature in β_0.

Example: $H_0 : \beta = \beta_0 \quad$ vs. $\quad H_1 : \beta \neq \beta_0$



Score-Test

- For the Score function $\mathbf{s}(\beta) = \mathbf{X}^\top \mathbf{D}(\beta)\mathbf{\Sigma}(\beta)^{-1}(\mathbf{y} - \boldsymbol{\mu})$, it is known:
  - $\mathbb{E}(\mathbf{s}(\hat{\beta})) = \mathbf{0}$
  - $\mathrm{Cov}(\mathbf{s}(\hat{\beta})) = \mathbf{F}(\hat{\beta})$
- If H_0 is true, $\mathbf{s}(\tilde{\beta})$ should behave similar $\implies$ regard the weighted distance of $\mathbf{s}(\tilde{\beta})$ towards $\mathbf{0}$

Score statistic

$$u = \mathbf{s}(\tilde{\beta})^\top \mathbf{F}(\tilde{\beta})^{-1}\mathbf{s}(\tilde{\beta}) \overset{a}{\sim} \chi^2_r$$

often hard to get

Testing hypotheses

Asymptotics of the three test statistics

For large n and under H_0:

$$\lambda, w, u \overset{a}{\sim} \chi^2_r \,,$$

i.e., reject H_0, if

$$\lambda, w, u > \chi^2_{1-\alpha;r}$$

!